CLOUD BASED VOTING SYSTEM

A SUMMER INTERNSHIP REPORT

Submitted by

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.

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BONAFIDE CERTIFICATE

This is to certify this Summer Internship project report **"Cloud Based Voting System"** is the Bonafide work of **GOKULNATH V (810023205116)** has satisfactory completed the course of **IT3711 – SUMMER INTERNSHIP** for **SEVENTH** semester, **B.TECH INFORMATION TECHNOLOGY** during the academic year **2026-2027.**

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Submitted for the practical examination held on _____.

ABSTRACT

The Cloud-Based Voting System is a web-based application designed to provide a secure, efficient, and convenient platform for conducting elections using cloud computing technology. The system aims to reduce the limitations of traditional voting methods by providing online voter registration, user authentication, OTP verification, candidate management, vote casting, vote counting, and result management.

The system is developed using HTML, CSS, and JavaScript for the frontend and Node.js with Express.js for the backend. Cloud services such as Amazon Web Services (AWS) can be used for application hosting, data storage, scalability, and reliable access. The system consists of two primary users: Voter and Administrator. Voters can register, log in, verify their identity through OTP, view candidates, and cast their votes. Administrators can manage voters, candidates, elections, and election results.

The proposed system provides centralized data management and reduces manual effort in the election process. Automated vote processing and result generation can help improve efficiency and reduce the time required for counting votes. The project demonstrates how cloud computing and web technologies can be integrated to develop a modern voting management application. With additional security, auditing, identity verification, and compliance mechanisms, the system can be further enhanced for more advanced election-management applications.

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1. INTRODUCTION

The Cloud-Based Voting System is a web-based application developed to provide a secure, reliable, and convenient method for conducting elections through cloud technology. The system is designed to simplify the voting process by allowing registered voters to authenticate themselves, verify their identity, cast their votes, and receive confirmation through an online platform.

Traditional voting systems generally require voters to visit a physical polling station. This process may require significant manpower, infrastructure, paper-based documentation, and manual vote counting. The proposed cloud-based voting system aims to reduce these limitations by providing a digital platform for voter registration, authentication, voting, vote storage, and result management.

The system uses HTML, CSS, and JavaScript for developing the interactive user interface. Node.js and Express.js can be used for backend development, while cloud services such as Amazon Web Services (AWS) can provide scalable and reliable infrastructure for storing and processing application data.

The system contains two major users: Voter and Administrator. The voter can register, log in, verify the account using OTP, view available candidates, and cast a vote. The administrator can manage elections, candidates, registered voters, and election results.

The main purpose of the project is to demonstrate how cloud computing can be applied to develop an efficient voting platform with improved accessibility, centralized data management, scalability, and faster result processing.

1.1 Objective of the Internship

The main objective of the internship is to understand the practical application of cloud computing and web development technologies by developing a Cloud-Based Voting System.

The major objectives are:

- To understand the fundamentals of cloud computing.
- To develop a web-based voting application.
- To understand frontend and backend development.
- To implement voter registration and authentication.
- To provide OTP-based user verification.
- To securely store voting-related information.
- To implement candidate and election management.
- To provide an automated vote-counting mechanism.

- To understand the role of AWS cloud services.
- To develop practical knowledge of Node.js and Express.js.
- To understand database connectivity and cloud-based data management.
- To analyze the advantages and limitations of an online voting system.

1.2 Objective of cloud based voting system

- To provide secure online voting.
- To authenticate eligible voters.
- To prevent duplicate voting.
- To securely store voting data.
- To manage voters and candidates.
- To provide easy vote casting.
- To automate vote counting.
- To generate accurate results.
- To reduce time and manual work

.2.COMPANY PROFILE



Fantasy Solution:

Trichy is a leading IT solutions and service provider offering software development, web designing, web hosting, and IT training services.

The organization specializes in real-time project development, application development & maintenance, ERP implementation & consulting, testing, performance engineering, and software localization.

Fantasy Solution has contributed to the formulation of several national IT standards and continues to deliver technology solutions in fields such as automotive electronics, smart devices, digital home products, and enterprise applications.

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Mission:

Fantasy Solution is committed to providing high quality software development services, secure access to information, improved communication between businesses, value-formoney solutions to customers, and meaningful growth opportunities for its employees.

Vision:

To be a leading software development and training organization by delivering innovative, high-quality, and industry-oriented technology solutions. The company aims to empower students and organizations through focused project guidance, practical learning, and professional software development services that enhance technical skills and support career growth. It strives to bridge the gap between academic learning and industry requirements by providing real-time project experience and continuous technological innovation.

3. DAILY WORK REPORT:

Domain: Cloud Computoing
Project: Cloud Based Voting System
Company: Fantasy IT Solutions

Day 1

The internship began with an introduction to the organization, internship objectives, and the training schedule. The basic concepts of cloud computing, its importance, applications, and benefits were introduced. An overview of the Cloud-Based Voting System project was also discussed.

Day 2

The basic characteristics of cloud computing were studied. Concepts such as on-demand services, resource sharing, scalability, availability, and pay-as-you-use models were discussed. Different applications of cloud computing were also explored..

Day 3

The major cloud service models, namely Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), were studied. Their features and applications in software development were discussed.

Day 4

Different cloud deployment models such as public cloud, private cloud, hybrid cloud, and community cloud were studied. The advantages of using cloud infrastructure for web applications were analyzed..

Day 5

An introduction to Amazon Web Services (AWS) was provided. Basic AWS services and their role in hosting, storage, computing, and application deployment were studied.

Day 6

The requirements of the Cloud-Based Voting System were analyzed. The major users, system requirements, functional requirements, and expected outputs were identified. The overall purpose and scope of the project were discussed.

Day 7

The overall architecture of the Cloud-Based Voting System was planned. The interaction between the voter, administrator, application server, database, and cloud environment was studied and documented.

Day 8

The basic concepts of web development were studied. HTML, CSS, and JavaScript were introduced for developing the frontend of the voting application..

Day 9

The basic structure of the voting system web pages was created using HTML. Pages such as Home, Registration, Login, and Dashboard were planned and developed.

Day 10

CSS was used to improve the appearance and layout of the application. Forms, buttons, navigation sections, and other interface components were designed to provide a simple and user-friendly experience.

Day 11

JavaScript concepts required for the project were studied. Form validation, button events, input handling, and interactive webpage functionality were implemented..

Day 12

Node.js was introduced for backend development. The role of server-side programming and the advantages of using Node.js for web applications were studied.

Day 13

Express.js was studied and used to create the backend structure of the application. Basic routes and server-side request handling were implemented.

Day 14

Database concepts required for the voting system were studied. The information required for voters, candidates, elections, and votes was identified and organized.

Day 15

The backend was connected with the database. Basic operations such as storing, retrieving, updating, and managing application data were studied and implemented.

Day 16

The voter registration module was developed. The registration form was created and the required voter information was collected and stored in the database.

Day 17

The user login functionality was implemented. Registered voters were allowed to enter their credentials and access the voting system after successful authentication.

Day 18

OTP-based verification was studied and integrated into the authentication process. The purpose of OTP verification in improving user authentication and security was understood.

Day 19

The candidate management module was developed. Candidate details such as candidate name, election information, and other required details were managed through the system.

Day 20

The election management module was implemented. The administrator can create and manage election details, including election name, candidates, and election status.Minor issues were identified and resolved.

Day 21

The voter dashboard was developed. After successful login and verification, voters can access available elections and view the required election information.

Day 22

The candidate selection interface was developed. The voter can view the available candidates and select the preferred candidate before submitting the vote.

Day 23

The vote casting functionality was implemented. The selected candidate information is submitted to the backend and stored in the database after successful validation.

Day 24

The system was designed to prevent the same voter from casting multiple votes in the same election. Voting status and validation mechanisms were implemented.

Day 25

The vote-counting functionality was developed. The system processes the stored voting information and calculates the number of votes received by each candidate.

Day 26

The administrator dashboard was developed. The admin can manage voters, candidates, elections, and view election-related information from a centralized interface.

Day 27

The role of cloud computing in hosting and managing the application was studied. AWS-based cloud infrastructure and cloud storage concepts were explored for the deployment of the voting system.

Day 28

The complete application was tested module by module. Registration, login, OTP verification, candidate selection, vote casting, admin operations, and result generation were tested. Errors were identified and corrected.

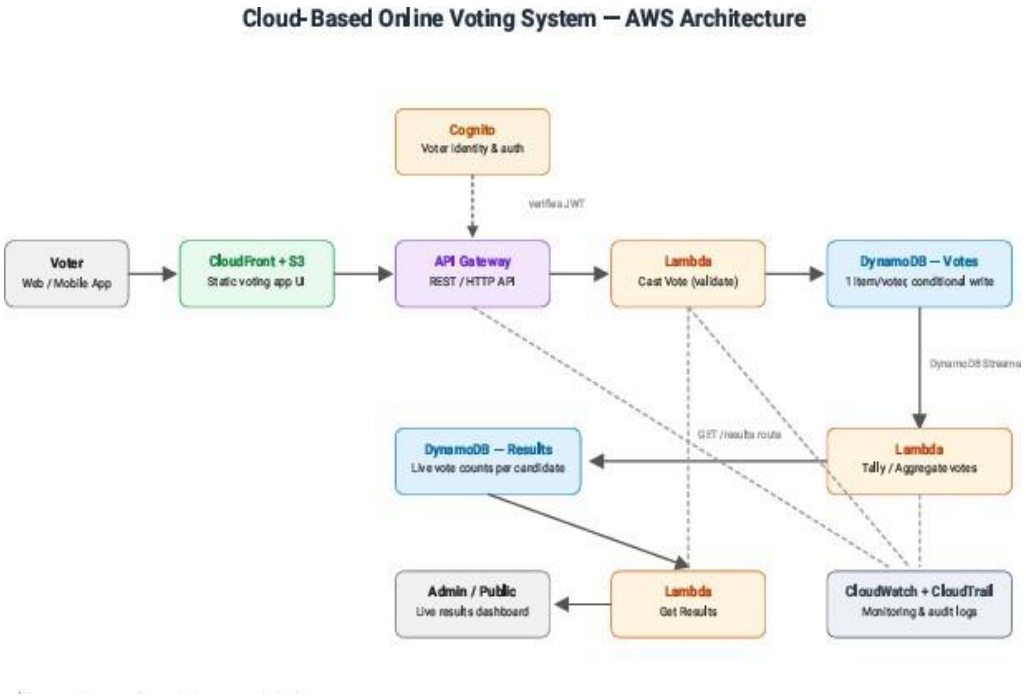
Day 29

All modules were integrated into the complete Cloud-Based Voting System. The project workflow, screenshots, outputs, implementation details, and technical documentation were prepared for the internship report.

Day 30

The completed Cloud-Based Voting System was reviewed and demonstrated. The overall project functionality, cloud computing concepts, technologies used, and learning outcomes were discussed. The internship activities and project documentation were finalized and submitted.

4. BLOCK DIAGRAM:



5. CLOUD BADES VOTING SYSTEM

The Cloud-Based Voting System is designed to provide a secure and efficient platform for conducting elections online. It allows voters to register, authenticate, cast their votes, and view election information. The voting data is securely stored in the cloud, while administrators can manage voters, candidates, elections, and results. The system reduces manual work, saves time, prevents duplicate voting, and provides faster and more accurate election results..

5.1 EXISTING SOLUTION

In the existing voting system, elections are mainly conducted using physical polling stations, paper ballots, or electronic voting machines. The process requires more manpower and infrastructure. Vote counting and result preparation can also take considerable time. Manual processes may lead to errors, paperwork, and difficulties in managing large amounts of election data..

5.2 PROPOSED SOLUTION

The proposed Cloud-Based Voting System provides an online platform for secure and convenient voting. Voters can register, authenticate themselves, and cast their votes electronically. Voting data is securely stored in the cloud, and votes can be counted automatically. Administrators can manage voters, candidates, elections, and results through a centralized system. This improves security, accuracy, accessibility, and efficiency while reducing manual work and processing time.

5.3 CAPABILITIES

The Cloud-Based Voting System provides the following main capabilities:

1. Voter Registration and Authentication – Allows voters to register and verify their identity before voting.
2. Secure Vote Casting – Enables authenticated voters to select and submit their vote securely.
3. Duplicate Vote Prevention – Prevents a voter from casting more than one vote.
4. Cloud Data Storage – Stores voter, candidate, and election data securely in the cloud.
5. Election Management – Allows administrators to manage candidates, voters, and election details.
6. Automatic Vote Counting – Counts votes automatically and reduces human errors.
7. Result Generation – Provides fast and accurate election results.
8. Data Security – Protects sensitive voting information using authentication and security techniques.

6. PROPOSED METHODS

The proposed Cloud-Based Voting System uses a set of secure and efficient methods to conduct the voting process digitally. The system focuses on voter authentication, secure vote casting, cloud storage, and automatic result generation. These methods help reduce manual work and improve the security, accuracy, and efficiency of the election process.

6.1 Voter Registration and Authentication

The system allows eligible voters to register using their personal details and voter ID. During login, the voter's credentials are verified. OTP verification can also be used to provide an additional layer of security.

6.2 Secure Vote Casting

After successful authentication, the voter can view the available candidates and select a preferred candidate. The system validates the voter before accepting the vote and prevents duplicate voting.

6.3 Cloud-Based Data Storage

Voter details, candidate information, election details, and voting data are stored securely using cloud storage. This provides centralized data management, easy accessibility, backup, and scalability.

6.4 Vote Encryption

Sensitive voting information is protected using encryption techniques before storage or transmission. This helps prevent unauthorized users from accessing or modifying confidential election data.

6.5 Automatic Vote Counting

The system automatically counts the votes received by each candidate. This reduces manual calculation, minimizes human errors, and helps generate election results quickly.

6.6 Result Management and Monitoring

The administrator can manage elections, candidates, and voters and monitor voting activities. After the election, the system displays the vote count and final results to authorized users.

Advantages of Proposed System

- Secure Voting – Protects voter and voting data through authentication and encryption.
- Fast Results – Automatically counts votes and generates results quickly.
- Easy Access – Allows eligible voters to cast their votes through an online system..

Disadvantages of Proposed Solution

- Internet Dependency – Requires a stable internet connection for voting.

- Security Risks – May face cyber attacks if proper security measures are not implemented.
- Technical Issues – Server failures or software problems may interrupt the voting process.

7. ALGORITHM

7.1 Voter Registration Algorithm

1. Start the system.
2. Enter voter details such as name, email, mobile number, and voter ID.
3. Validate the entered details.
4. Check whether the voter is already registered.
5. If valid, create the voter account.
6. Store the voter details securely in the cloud database.
7. Display successful registration.
8. Stop.

7.2 Voter Authentication Algorithm

1. Enter username and password.
2. Verify the login credentials.
3. Send OTP for additional verification.
4. Verify the OTP.
5. If authentication is successful, allow access to the voting page.
6. Otherwise, display an authentication error.

7.3 Vote Casting Algorithm

1. Authenticate the voter.
2. Display the available election and candidates.
3. Allow the voter to select one candidate.
4. Validate the voter eligibility.
5. Encrypt the vote.
6. Store the encrypted vote in the cloud database.
7. Mark the voter as having voted.
8. Display vote submission confirmation.

7.4 Vote Counting Algorithm

1. Retrieve the stored votes from the cloud database.
2. Decrypt the votes using the authorized key.
3. Count the votes for each candidate.
4. Compare the vote counts.
5. Determine the candidate with the highest number of votes.
6. Store and display the election results.

8. CODE IMPLEMENTATION:

```
<!DOCTYPE html>
<html>
<head>
   <title>Cloud Based Voting System</title>
</head>
<body>

<h2>Cloud Based Voting System</h2>

<h3>Voter Login</h3>

<input type="text" id="voterId" placeholder="Voter ID">
<input type="password" id="password" placeholder="Password">
<button onclick="login()">Login</button>

<h3>Cast Your Vote</h3>

<select id="candidate">
   <option value="">Select Candidate</option>
   <option value="Candidate A">Candidate A</option>
   <option value="Candidate B">Candidate B</option>
   <option value="Candidate C">Candidate C</option>
</select>

<button onclick="vote()">Submit Vote</button>

<h3>Election Result</h3>
<button onclick="showResult()">View Result</button>

<p id="message"></p>

<script>

let loginStatus = false;
let voted = false;

let votes = {
   "Candidate A": 0,
```

```
    "Candidate B": 0,
    "Candidate C": 0
};

function login() {

    let id = document.getElementById("voterId").value;
    let pass = document.getElementById("password").value;

    if (id != "" && pass != "") {
        loginStatus = true;
        document.getElementById("message").innerHTML =
            "Login Successful";
    }
    else {
        document.getElementById("message").innerHTML =
            "Enter valid login details";
    }
}

function vote() {

    if (!loginStatus) {
        document.getElementById("message").innerHTML =
            "Please Login First";
        return;
    }

    if (voted) {
        document.getElementById("message").innerHTML =
            "You have already voted";
        return;
    }

    let candidate =
        document.getElementById("candidate").value;

    if (candidate == "") {
        document.getElementById("message").innerHTML =
            "Select a Candidate";
```

```
        return;
    }

    votes[candidate]++;
    voted = true;

    document.getElementById("message").innerHTML =
        "Vote Submitted Successfully";
}

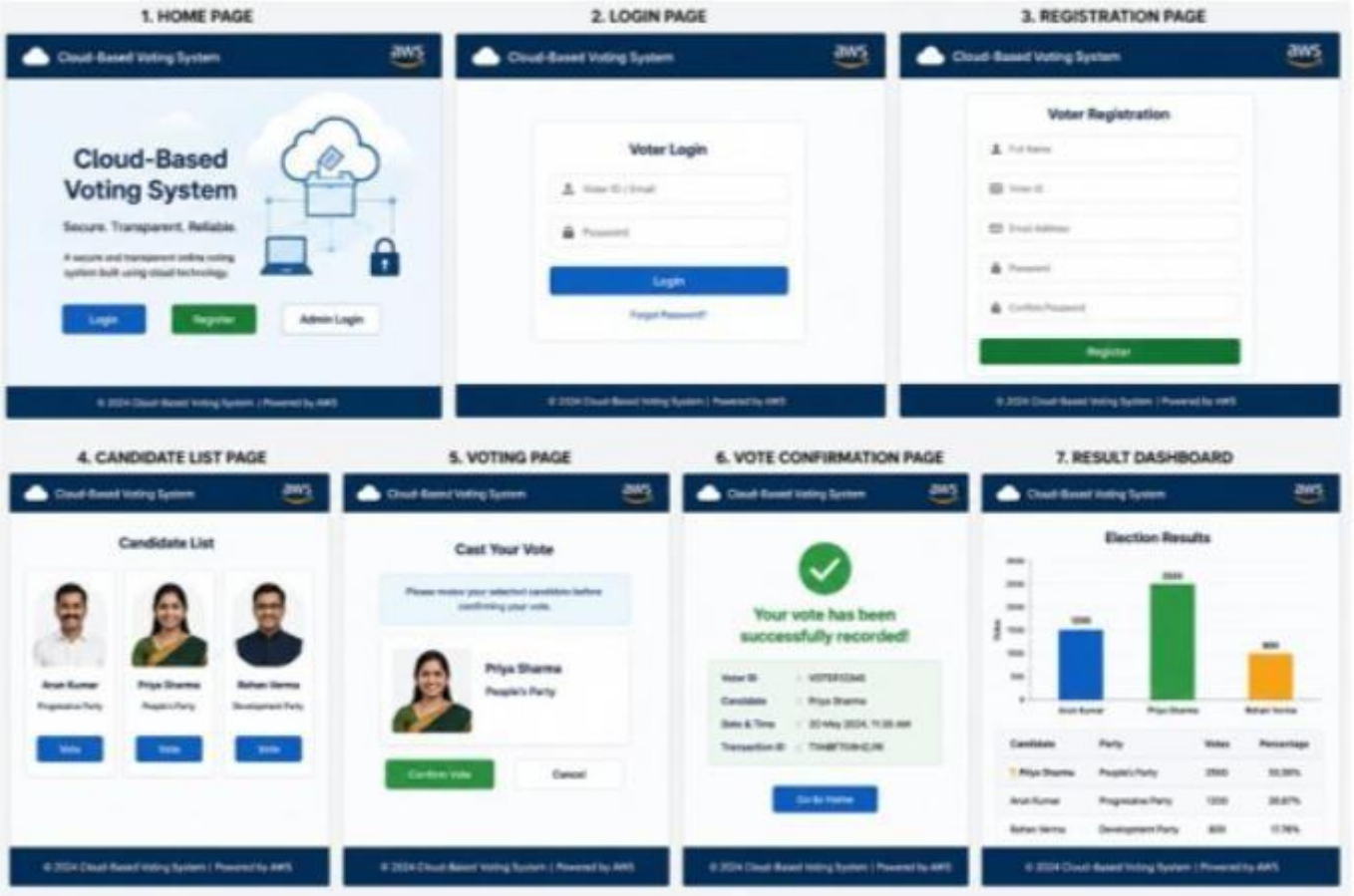
function showResult() {

    document.getElementById("message").innerHTML =
        "Candidate A: " + votes["Candidate A"] + " Votes<br>" +
        "Candidate B: " + votes["Candidate B"] + " Votes<br>" +
        "Candidate C: " + votes["Candidate C"] + " Votes";
}

</script>

</body>
</html>
```

9. SAMPLE OUTPUT:



10. ANALYSIS:

The Cloud-Based Voting System provides a secure and efficient alternative to traditional voting methods. The system reduces manual work and makes the voting process faster. Authentication helps ensure that only eligible voters can participate, while encryption protects sensitive voting information. Cloud storage provides centralized management and easy access to election data. Automatic vote counting reduces human errors and allows results to be generated quickly. Overall, the proposed system improves the security, efficiency, accessibility, and reliability of the voting process..

11. CONCLUSION:

The Cloud-Based Voting System provides a secure and convenient platform for conducting elections electronically. The system uses voter authentication, OTP verification, secure cloud storage, encryption, and automatic vote counting to improve the reliability of the election process. It reduces paperwork, manual effort, and the time required for vote counting. The proposed system demonstrates how cloud computing can be used to develop an efficient and secure digital voting solution.

12. FUTURE WORKS

- Biometric Authentication – Fingerprint or facial recognition can be added for stronger voter verification.
- Blockchain Integration – Blockchain technology can be used to improve vote transparency and prevent unauthorized modification.
- Mobile Application – A dedicated Android or iOS application can be developed for easier access.
- Advanced Security – Multi-factor authentication and advanced threat detection can be added to improve system security.
- Real-Time Monitoring – Administrators can monitor election activities and voting statistics in real time.

13. REFERENCES

1. AWS Documentation – Cloud Computing and Cloud Storage.
2. Node.js Documentation – Server-side JavaScript development.
3. Express.js Documentation – Web application framework.
4. MongoDB Documentation – Database management and storage.
5. NIST – Guidelines and publications related to cybersecurity and authentication.
6. OWASP – Web application security guidelines.